QUANT

Permutation, Combination & Probability

Assignment 03

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P & C Assignments 03

1. In how many ways 5 Indians and 4 Englishmen can be seated around a round table if no two Englishmen sit together?

1. $4! \times 5!$ 2. $5! \times 5!$ 3. $4! \times 4!$ 4. $4! \times 3!$

2. A round table conference is to be held between 20 delegates of 20 countries. In how many ways can they be seated if two particular delegates are always to sit together?

1. 18! 2. $19! \times 2!$ 3. 20! 4. $18! \times 2!$

3. A man has to take 8 children to a zoo. He takes them to the zoo three at a time, as often as he can, without taking the same 3 children together more than once. How many times will he have to go to zoo? How many times does a particular child go?

1. 42, 21 2. 21, 56 3. 56, 21 4. 42, 56

4. Find the number of diagonals of a polygon which has *m* sides.

1. $\frac{m(m-3)}{2}$ 2. $\frac{m(m-1)}{2}$ 3. $\frac{m(m-2)}{2}$ 4. $\frac{m^2}{2}$

5. In how many ways can 12 different things be divided equally among 3 persons

1. $\frac{12!}{(4!)^3 \times 3!}$ 2. $\frac{12! \times 3!}{(4!)^3}$ 3. $\frac{12!}{(3!)^4}$ 4. $\frac{12!}{(4!)^3}$

6. In how many ways can 12 things be divided in three groups having 4 things each?

1. $\frac{12!}{(4!)^3 3!}$ 2. $\frac{12!}{(4!)^3}$ 3. $\frac{12! \times 3!}{(4!)^3}$ 4. $\frac{12!}{(3!)^4}$

7. The number of straight lines that can be drawn out of 10 points of which 7 are collinear is

1. 24 2. 23 3. 25 4. 21

8. The number of five–digit telephone numbers having at least one of their digits repeated is

1. 90,000 2. 1,00,000 3. 30,240 4. 69,760

9. You are given 5 line segments to lengths 2, 3, 4, 5, 6 units. Then the number of triangles that can be formed by joining these lines is

1. ${}^5C_3$ 2. ${}^5C_3 - 3$ 3. ${}^5C_3 - 2$ 4. ${}^5C_3 - 1$.

10. If eight persons are to address a meeting, then the number of ways in which a specified speaker is to speak before another specified speaker is

1. 2,520 2. 20,160 3. 40,320 4. None of these

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Answer key & Explanations

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1.	1	Since there is no. restriction on Indians. $\therefore$ First we arrange 5 Indians, this can be done around a table in =4! ways. If the Englishman sit at the places in between in then no two of these will be seated together. $\therefore$ There are 5 places for 4 Englishman, they can be arranged in ${}^5P_4$ ways; $\therefore$ Required no. $4! \times {}^5P_4 = 4! \times 5!$
2.	4	Treating 2 particular delegates who are to sit together as one person, we have only 19 persons. These 19 persons can be seated at a round table in 18! Ways. But 2 particular persons can be arranged among themselves in 2! ways. $\therefore$ Required no = $18! \times 2!$.
3.	3	No. of children = 8 No. of children to be taken at a time =3 ; Out of 8 children 3 can be selected in ${}^8C_3$ ways. Hence man will have to go to zoo ${}^8C_3$ ways $= \frac{8!}{3!5!}$ =56times; 2nd part – No. of selections of 3 children out of 8 children including a particular child $= {}^1C_1 \times {}^7C_2 = \frac{7!}{2!5!} = 21$ times.
4.	1	A polygon of '*m*' sides will have '*m*' vertices. When any two vertices of the polygon are joined, either a diagonal or a side of the polygon will be formed. Now number of selection of m different points taken 2 at a time = ${}^mC_2$. $\therefore$ ${}^mC_2$= No. of sides + No. of Diagonals. $\therefore$ No. of Diagonals = ${}^mC_2$-m= $\frac{m(m-3)}{2}$
5.	4	Each person will get 4 things. Now first person can be given 4 things out of 12 things in ${}^{12}C_4$ ways. Second person can be given 4 thins out of 8 things in ${}^8C_4$ ways. Third person can be given 4 things out of remaining 4 things in ${}^4C_4$ ways ; $\therefore$ Required number = ${}^{12}C_4 \times {}^8C_4 \times {}^4C_4 = \frac{12!}{(4!)^3}$
6.	1	No. of ways in which 12 different things can be divided among 3 sets $= \frac{{}^{12}C_4 \times {}^8C_4 \times {}^4C_4}{3!} = \frac{12!}{(4!)^3 3!}$
7.	3	The required number of lines =1+${}^{10}C_2$ - ${}^7C_2$=46-21=25
8.	4	The number of five–digit telephone numbers which can be formed using the digits of 0, 1, 2, …., 9 is 10^5. The number of five digit telephone numbers which have none of their digits repeated is ${}^{10}P_5$=30240. Hence the required number = 10^5 -30240=69760
9.	2	We know that in any triangle the sum of two sides is always greater than the third side. $\because$ the triangle will not be formed if we select segments of lengths (2, 3, 5) , (2, 3, 6) and (2, 4, 6). Hence no. of triangles formed =${}^5C_3$-3
10.	2	Let A, B be the corresponding speakers. Without any restrictions the eight persons can be arranged among themselves in 8! Ways, but the number of ways in which A speaks before B and the number of ways in which B speaks before A make up 8! Also number of ways in which A speaks before B is exactly same as the number of ways in which B speaks before A.

		$\therefore$ the required no. of ways = $\frac{1}{2}(8!)$=20160.